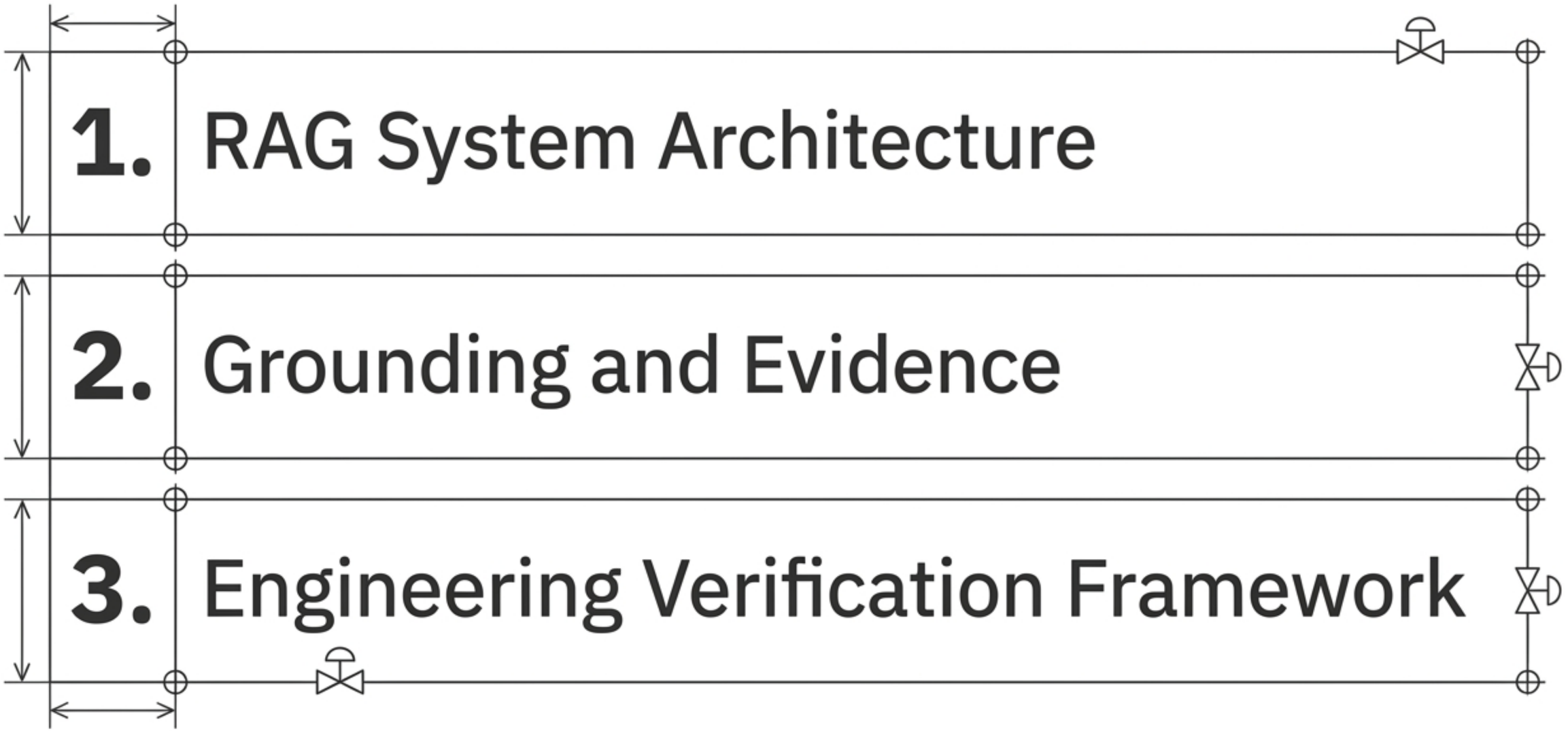
MODULE 14, LESSON 2

AI 101 for Mechanical Engineers

Retrieval supplies evidence at inference time

Gemini Notebook

Course structural outline



Gemini Notebook

1. RAG System Architecture

2. Grounding and Evidence

3. Engineering Verification Framework

1. RAG System Architecture

Defines the operational pipeline and mandatory parameter constraints.

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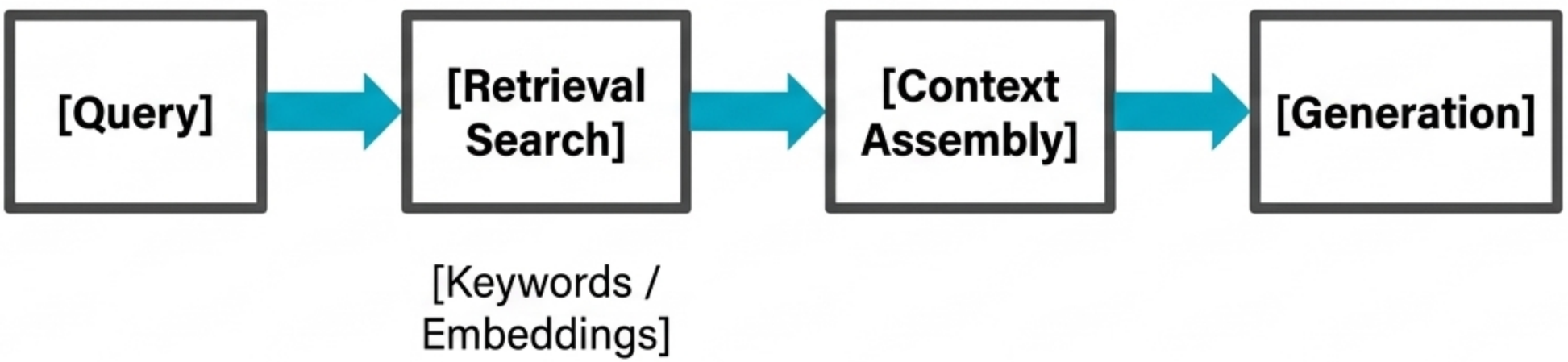
1. RAG System Architecture

2. Grounding and Evidence

3. Engineering Verification Framework

Architecture systematically constructs context before generating outputs

- System receives input engineering query.
- Retrieval searches collection via keywords or embeddings.
- System adds retrieved passages to model context.
- Model generates answer using assembled context.



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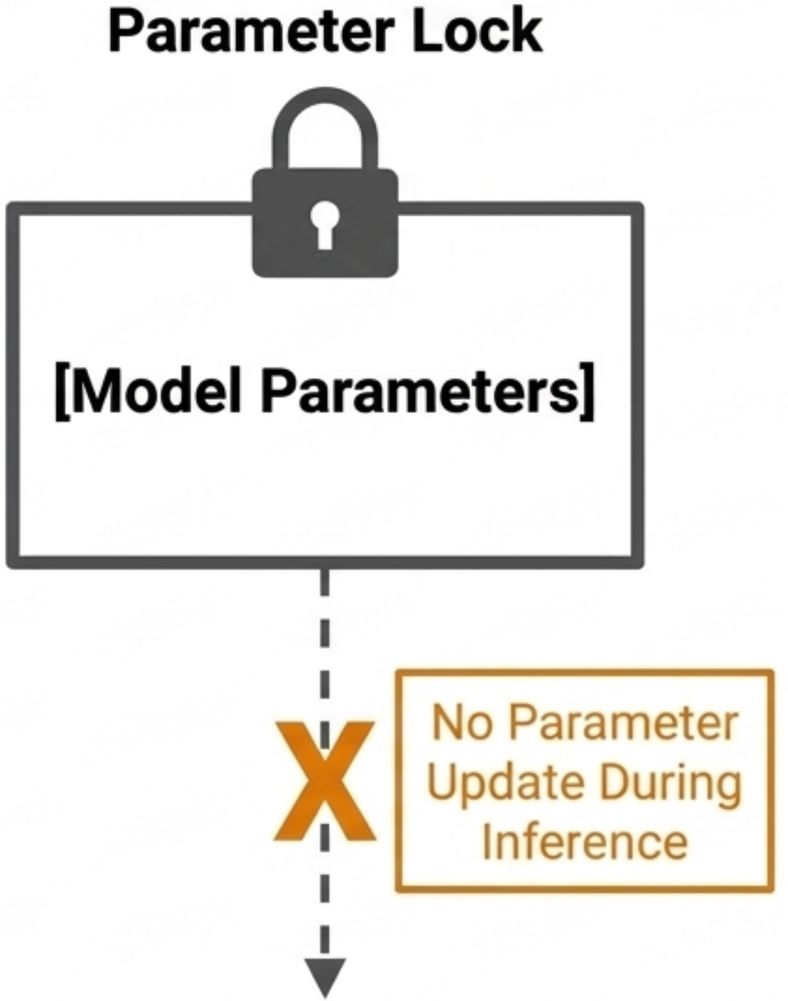
1. RAG System Architecture

2. Grounding and Evidence

3. Engineering Verification Framework

Model weights remain strictly locked during retrieval operations

- RAG supplies external evidence at inference time.
- System does not inherently update model parameters.
- Model weights remain completely static during retrieval.
- Retraining weights requires a separate engineering operation.



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2. Grounding and Evidence

Defines output connection structures and inherent systemic limitations.

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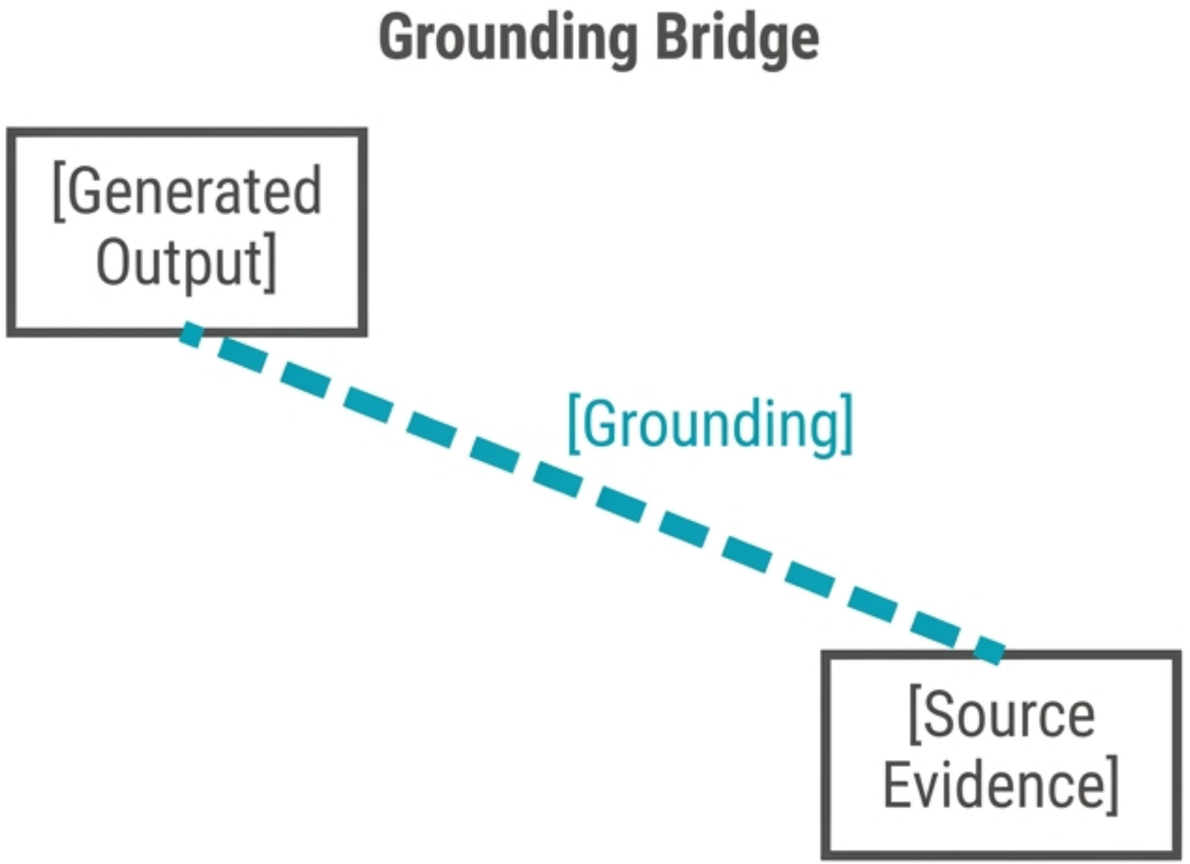
1. RAG System Architecture

2. Grounding and Evidence

3. Engineering Verification Framework

Grounding establishes a structural link to retrieved evidence

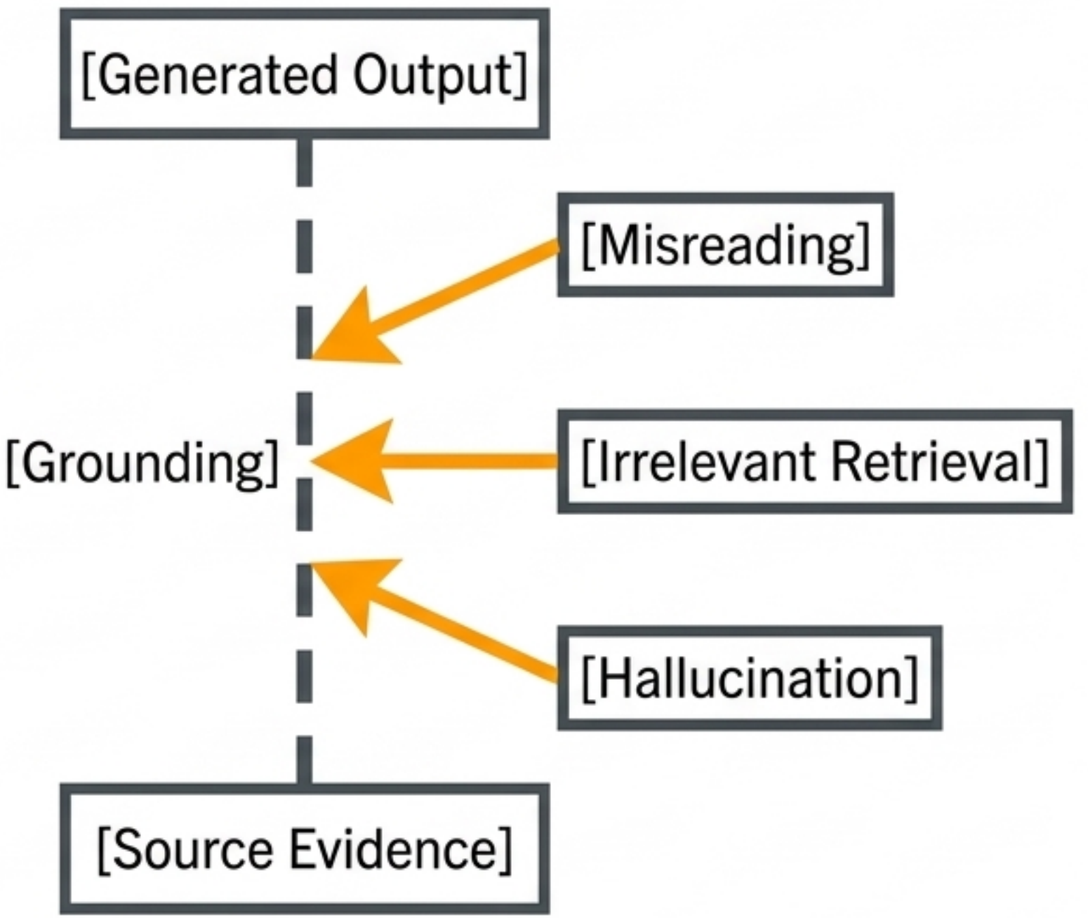
- Grounding **connects generated output to supplied evidence**.
- Architecture **improves** capability for **manual source checking**.
- Example: Linking **output to actual datasheet passages**.



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Grounding structures do not eliminate inherent system failure modes

- System remains vulnerable to misreading retrieved data.
- Retrieval algorithms may pull irrelevant technical documents.
- Model hallucination occurs despite grounded evidence connections.
- Similar terminology does not guarantee accurate evidence.



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3. Engineering Verification Framework

Establishes strict manual checking protocols for automated AI outputs.

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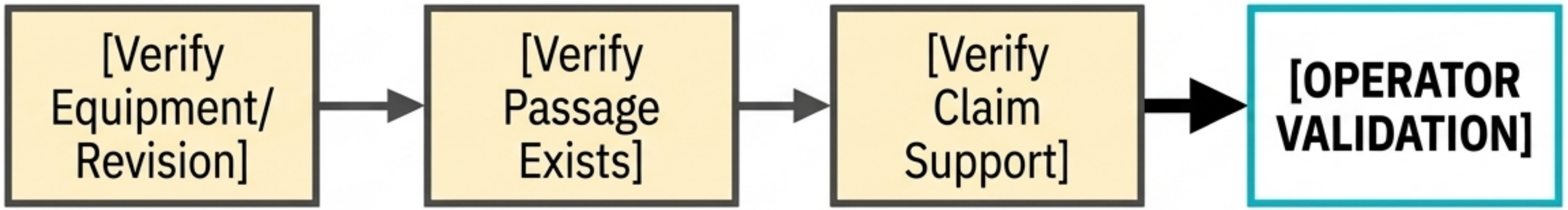
Automated generation stages mandate parallel manual verification steps

RAG Stage (System)	Evidence Check ([OPERATOR VALIDATION])
Retrieve	Confirm correct equipment and document
Add source context	Confirm correct revision and passage
Generate	Confirm claim supported by passage
Review	Original source physically opened

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Operators must execute a sequential three-gate logic check

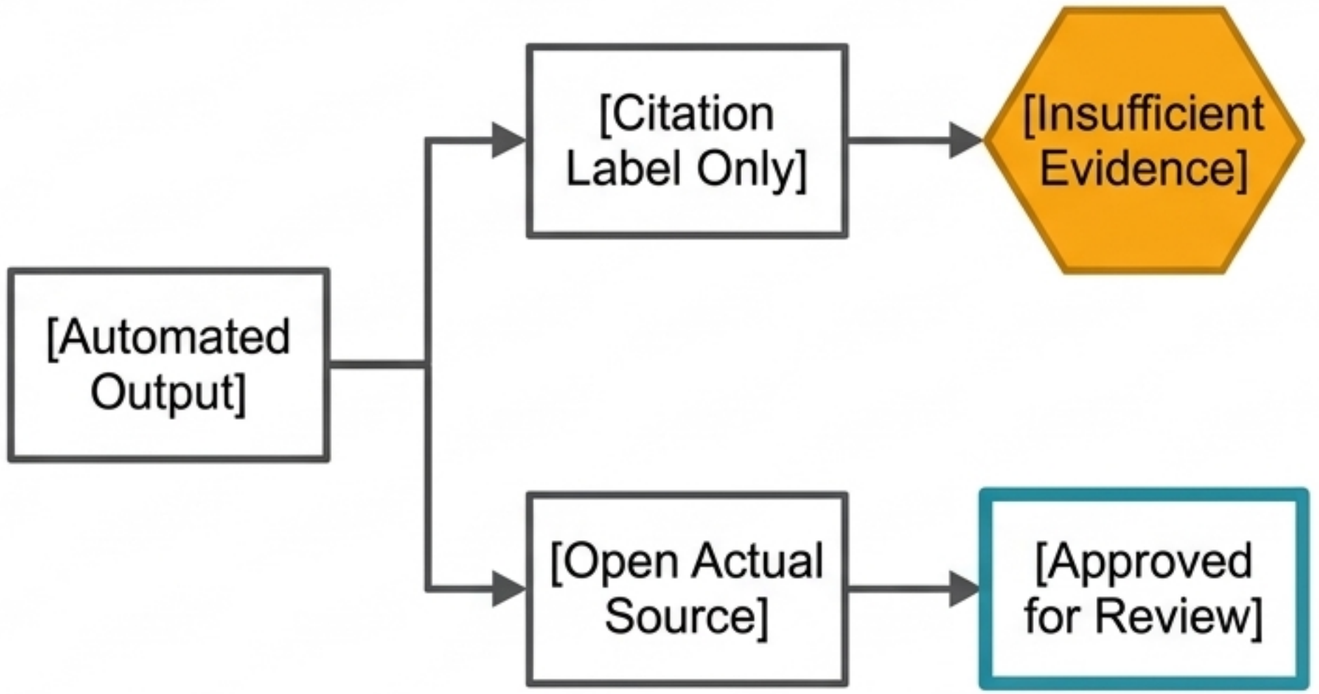
- Confirm precise equipment matches target machine.
- Verify cited passage actually exists in revision.
- Confirm passage technically supports claimed operating limit.
- Confident output wording never substitutes actual support.
- Confident output wording never substitutes actual support.



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Citation labels never constitute operational proof without physical verification

- Output citation labels require immediate physical authentication.
- Never treat generated UI labels as proof.
- Operator must open original approved manual physically.



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Key Points

- **Architecture**: RAG retrieves context without altering underlying model parameters.
- **Evidence**: Grounding connects outputs but retains critical failure modes.
- **Verification**: Operators must manually verify source existence and claim support.

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